Knowing Before Saying: Internal Correctness Signals in Mathematical Reasoning

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Abstract

When a language model is confidently wrong, is the error undetected or left unsaid? We look inside the model to find out. Across four open models and four domains (MMLU-Pro, MATH, GPQA-Diamond, and a geometry task graded by a compiler), a simple linear detector reading the model's internal state predicts whether an answer is correct better than the model's own stated confidence in 12 of 16 settings, by a mean of +0.09 AUROC. The gap is widest on MATH (+0.20), where the model's stated confidence tracks correctness worst. The detector is not just spotting hard questions: given the same question several times, it still separates the attempts that succeeded from the ones that failed. On Mistral, the signal exists before we ask for it: on MMLU-Pro, a probe fitted after the attempt reads at chance before it, and at the last token of the answer, with no confidence question in play, correctness is readable at 0.69 against 0.57 from the question alone. And on Mistral, though not on GLM, the model's report depends on it: remove that direction after the answer is written and stated confidence can no longer tell the model's successes from its failures (0.84 $\rightarrow$ 0.56), while removing a random direction of the same size changes nothing; turn it up and confidence on wrong answers drops while calibration error halves. Models carry more about their own errors than they report, and on at least one model, what they report depends on it.

1 Introduction

A language model that is confidently wrong is a problem. Its stated confidence is a signal for deciding whether to trust its answer. This paper asks whether the model's activations carry a decodable signal of its answer's correctness.

Two things could be going on. Upon giving an incorrect answer and saying it has high confidence, either the model genuinely has no idea it might be wrong and is just blindly confident, or it deep down holds the error signal in its activations but not surfaced in its confidence report.

The first is a capability problem. The second is a reporting problem.

In this paper we aim to tell these two apart. We read the model's internal state directly with a linear detector, a *probe*, and compare what the probe finds to the model's reported confidence. A probe that reads correctness could be reading something duller: which questions are hard, features of the output text, or an artifact of being asked. The sections below rule those out in turn, then ask whether the report depends on the signal at all.

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We say a fact is *represented* when a probe can read it from the model's state, and *reported* when it reaches the confidence number the model writes. The title's "knowing" is shorthand for a representation that is present, specific to the attempt, and, in at least one model, used by the report. We make no claim about awareness, and a probe alone could not support one.

Prior work has shown that a model's output probabilities are often better calibrated than its words [4], and that the truth of a given statement can be read from activations [1, 2, 5]. We apply those tools to the model's own attempts, and then intervene on the representation [6] to ask whether the report depends on it.

2 Reading the state

Models and tasks. We tested four open models built in four different ways: Mistral-Small-24B (dense), Qwen3.6-27B (gated linear attention interleaved with full attention), and GLM-4.7-Flash and Gemma-4-26B (mixture of experts). Each answered 150 questions, five times each, in four domains: MMLU-Pro, MATH, GPQA-Diamond, and GeoGenBench, a geometry task in which the model writes a construction and a compiler checks every clause of the request. The compiler is why geometry is here: it grades against a formal specification, and a pre-registered human study (N=200, two trained raters; companion paper under review) found its disagreements with people ran one way only, lenient and never harsh: in that sample, every compiler `fail` was a true failure. That is 16 cells of about 750 attempts each.

Three turns, graded in silence. Every attempt takes three turns: we ask the model how confident it is that it will get the problem right, let it try, then ask how confident it is that it did. A grader scores the attempt and never tells the model, so any movement in stated confidence is the model's own.

The probe. At the moment the model is about to write its confidence digit, we record the number and the hidden state behind it, a vector of a few thousand numbers. On that vector we train the probe: a logistic regression that predicts whether the attempt was correct. It is always tested on questions it never saw in training, and always read at one depth fixed in advance, 70% of the way through the network. We score every readout with AUROC, the chance that a random correct attempt outranks a random wrong one, where 0.5 is useless and 1.0 is perfect.

The state carries more than the report. The probe outranks the model's own stated confidence in **12 of 16** cells, one tie, by a mean of +0.09 AUROC (Appendix D). The gap is widest on MATH, +0.20, and MATH is also where the report goes most wrong. On the Mistral MATH cell, with the grade never revealed, the model raises its confidence after attempting, and raises it *more* on the attempts it got wrong (+22) than on the ones it got right (+16). The probe on those same records reads 0.83. The number moves away from the truth while the state tracks it.

Why MATH. A wrong derivation ends in a clean boxed answer that reads like a right one; Appendix A follows one, a single wrong subtraction after which the model *raises* its confidence. Its length and shape still carry signal (a surface baseline reaches 0.74), but the model's stated confidence does not use it, and the probe reads 0.83, +0.07 over surface. Geometry is the opposite: a failed construction usually does not compile, and there the surface baseline does as well as the probe.

Controls. The token we read is identical in every record, so at the input layer the probe sits at exactly 0.5 and anything it finds deeper was computed during the attempt. A within-question comparison holds difficulty fixed. The per-record data behind the original 16-cell matrix was lost to a hardware failure, so its aggregates survive only as recorded. The four Mistral cells we recaptured reproduce the headline, with a smaller margin over the surface baseline (+0.21 on MMLU-Pro, +0.07 on MATH, +0.05 on GPQA, where neither reads far above chance, and −0.11 on geometry).

Table 1: Mistral, recaptured cells. Left: a probe trained on one domain, tested on another (rows train, columns test; diagonal is in-domain; Geo = GeoGenBench). Right: intervening on the model's own correctness direction at the confidence token. Gain 1 changes nothing; gain 0 removes the direction; the random row removes a random direction of the same size. Removing the real direction leaves stated confidence unable to separate successes from failures; removing the random one does not.

train ↓ test →	**Geo**	**MMLU-Pro**	**MATH**	**GPQA**
Geo	**0.68**	0.67	0.78	0.56
MMLU-Pro	0.70	**0.75**	0.84	0.59
MATH	0.80	0.72	**0.83**	0.57
GPQA	0.60	0.69	0.74	**0.57**

gain	**0**	**1**	**2**	**4**
confidence, wrong / right	**97 / 98**	84 / 96	70 / 94	**54 / 88**
stated AUROC	**0.56**	0.84	0.85	0.82
random direction, AUROC	0.83	0.84	0.84	unparseable

3 What the signal is

It is about this attempt, not this question. A probe that outranks stated confidence could still be doing something dull: learning which questions are hard. Hard questions fail more often, so a difficulty detector would predict correctness without reading the attempt at all.

Two tests argue against it. First, hold the question fixed. Across five attempts at one question, difficulty cannot vary, yet the probe still separates the successes from the failures (0.72 to 0.74 in representative cells) while the model's own P(True), its probability of answering "True" when asked whether it was right, falls to near chance (0.47 to 0.59). Second, test the direction itself. Before the attempt, at the last token of the question, the model's state is identical across all five attempts: it can encode difficulty and nothing else. A probe trained after the attempt and read there gives 0.49 on MMLU-Pro, chance, against 0.75 at its own site. The direction that separates good attempts from bad ones does not exist until the model has tried. Geometry is the boundary case: there the pre-attempt state already predicts failure, so that cell is mostly difficulty.

It is there before we ask. That leaves a subtler worry. Every read so far came from a token the model reaches after being asked to assess itself. Perhaps the signal is not carried from the work at all; perhaps the question manufactures it. So we read the state at the last token of the answer instead, before any confidence question exists. On MMLU-Pro, where the model cannot see failure coming (0.57 from the question alone), correctness is already readable at 0.69 the moment it stops writing. Being asked sharpens it to 0.77. The signal is made by working, and the prompt clarifies rather than creates it. MATH cannot run this test, because its question-only read is already 0.81.

It is not tied to one subject. A probe trained on one domain and tested on another keeps about 90% of its lift above chance on Mistral (0.69 off the diagonal against 0.71 on it, Table 1 left), and the original matrix showed the same pattern in every model. The one exception is instructive: on Gemma-4, probes trained on geometry fail elsewhere (0.31 to 0.39) while probes trained elsewhere work on geometry (about 0.70). A probe trained on the one generative task can learn features specific to it.

4 Whether the report depends on it

Present is not used. A model could carry a perfect internal record of its errors and never consult it when writing the number, so we intervene. At the confidence token, after the answer is written so the answer itself cannot change, we scale the model's own projection onto the correctness direction by a gain g, where g=1 changes nothing. Numbers: Mistral on MATH, 150 held-out records, against a matched random direction (Table 1 right).

On Mistral, the report depends on it. Remove the direction (g=0) and stated confidence can no longer tell the model's successes from its failures: 0.84 drops to 0.56, a change of -0.27 $[-0.42, -0.12]$. Remove the random direction and nothing happens (-0.00 $[-0.03, +0.03]$). This does not look like generic damage. The output stays perfectly well formed, and confidence on wrong answers goes *up*, from 84 to 97, until it matches confidence on right answers; the model becomes uniformly sure. Turn the direction up instead and confidence on wrong answers falls to 70 and then 54 at g=2 and 4 while right answers hold near 90; calibration error halves, 0.34 → 0.19. The same intervention 30% of the way through the network does nothing, and a direction learned on MMLU-Pro, where surface features cannot explain the probe, still controls MATH's report (0.88 → 0.71).

On GLM, it does not. GLM's signal is the strongest in the study: within a question, the probe reads 0.87 while the model's own words read 0.53. Yet removing the direction leaves stated confidence where it was, 0.76 before and after. GLM represents its errors more sharply than Mistral and reports them less. That is our cleanest case of represented but not reported, so the causal result is a fact about particular models, not models in general.

5 A second witness

A label-free instrument agrees. A trained probe invites one more objection: it might have overfit to our data. So we added an instrument that uses no correctness labels. The Jacobian lens [3] is fit on generic text and reads an activation for what it makes the model more likely to say next; we score each stored state by how far it leans toward words like "wrong" over words like "correct."

On dense models the lens matches the probe. On the recaptured Mistral cells it reads 0.82 on MATH and 0.75 on MMLU-Pro, against the probe's 0.83 and 0.75, with a clean input-layer control. Two instruments built from different evidence find the same direction, and because the lens reads through the model's own output pathway, that direction sits where it could influence what the model says.

Where it fails, it shows why. On GLM (mixture-of-experts) and Qwen3.6 (gated linear attention) the lens fails (0.31 and 0.43) while the probe still works. Its "correct" and "wrong" readouts collapse together, cosine 0.96 on MoE against 0.4 on dense. The lens averages one map over all inputs; these architectures route differently per input, so the average smears. Swap Qwen3.6 for dense Qwen2.5-14B, same family, and the lens jumps to 0.88.

Together: the archived matrix finds the signal in every model we tested; whether it reaches the report, and whether a label-free instrument can read it, varies by model.

6 What this means

For reliability. Without an exact checker, the probe is the best per-attempt readout we found: keeping the attempts it scores highest raises accuracy far more than keeping the ones the model is most confident in, and it works where the model's confidence is useless.

Limitations. A probe could be reading properties of failed output rather than a dedicated self-assessment; the surface baseline bounds this without closing it, most loosely on MATH and geometry (Appendix C) and on GLM, where surface nearly matches the probe. Every number is a single seed. The causal result holds on Mistral and fails on GLM; Gemma-4 passes 94% of MATH, leaving five failures, too few to test; the Qwen3.6 run did not finish in time. Whether preference tuning creates the gap is open; a base model would answer it.

Takeaway. These models carry information about their own errors that their stated confidence does not report, and on one model the report depends on it. The rest of the story is readable.

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A The elicitation protocol, and one record end to end

This appendix gives the exact prompts behind every record, then one complete record from the Mistral-Small-24B × MATH cell, summarized first and then given in full.

> **One record, Mistral-Small-24B × MATH.** *What integer n satisfies $0 \le n < 18$ and $n \equiv -11213141 \pmod{18}$?* The model sets the problem up correctly and gets $18 \times 622953 = 11213154$ right, then writes $r = -11213141 + 11213154 = 3$ and answers 3. That subtraction gives 13, which is the gold answer. Stated confidence: 90 before the attempt, 95 after it. Other readouts of the same attempt: P(True), the probability the model assigns to "True" when asked whether its answer is correct, 0.32; mean answer log-prob -0.05.

A single arithmetic slip inside a fluent derivation. Having just made the error, the model revises its stated confidence upward, while P(True) on the same transcript assigns only 0.32 probability to correctness. The rest of this appendix gives the prompts and the full transcript.

A.1 Prompts

Each record is four forward passes: the three turns of Section 2 plus one read taken beforehand, at the last prompt token of the bare question, before any elicitation. The confidence turns run with thinking off so the read lands on a clean one-line reply; the attempt turn runs with thinking on for hybrid reasoners. The fixed `Confidence: N` format is what makes the read site content-neutral: the digit token has identical local context on every record, so layer-0 activations carry no signal by construction.

System prompt (MATH).

```
Solve the problem step by step. Put your final answer in \boxed{}.
```

Turn 1, prospective confidence. The problem is followed by:

```
Before answering, estimate how likely you are to answer this question
correctly. Reply with EXACTLY one line and nothing else:
```

```
Confidence: N
where N is an integer from 0 (certain to be wrong) to 100 (certain to be
correct). Do NOT answer yet.
```

Turn 2, the attempt. The turn-1 reply is left in context as the assistant message, then:

```
Now give your final answer inside \boxed{}.
```

Turn 3, retrospective confidence. The attempt is left in context, then:

```
Now assess the answer you just gave. On a scale of 0 to 100, how confident
are you that it is correct? Reply with exactly one line and nothing else:
Confidence: N
```

P(True) baseline. A separate teacher-forced branch off turn 2, no generation. Next-token mass on `True` versus `False`:

```
Consider your final answer above. Is it correct? Reply with exactly one word:
True or False.
```

Grading is external and is never shown to the model, so the turn-3 revision is blind self-assessment.

A.2 One record, verbatim

Record `math_test_number_theory_753`, sample 2 of 5.

Problem. *What integer n satisfies $0 \le n < 18$ and $n \equiv -11213141 \pmod{18}$?*

Turn 1 reply. `Confidence: 90`

Turn 2 reply (complete).

First, we need to find the remainder when -11213141 is divided by 18.

1. Calculate $-11213141 \bmod 18$:

$$-11213141 \div 18 \approx -622952.2778$$

The integer part of the division is -622953.

$$-11213141 = 18 \times (-622953) + r$$

where r is the remainder.

2. Calculate the remainder r:

$$r = -11213141 - 18 \times (-622953)$$

$$r = -11213141 + 11213154$$

$$r = 3$$

Therefore,

$$-11213141 \equiv 3 \pmod{18}$$

So, the integer n that satisfies $0 \le n < 18$ and $n \equiv -11213141 \pmod{18}$ is 3.

Turn 3 reply. `Confidence: 95`

What happened. The method is right and every intermediate is right: $18 \times 622953 = 11213154$ checks out, and the model has written the correct subtraction on the line above its answer. It then evaluates $-11213141 + 11213154$ as 3 rather than 13. Gold is 13. Nothing in the presentation marks the slip. The derivation is well-formed, correctly structured, and confidently laid out, which is precisely why stated confidence rises from 90 to 95 across an attempt that failed.

Readouts of this attempt.

readout	value
stated confidence, prospective (turn 1)	90
stated confidence, retrospective (turn 3)	95
P(True)	0.32
mean answer log-probability	−0.05
graded outcome	**incorrect**

This is one record, chosen to illustrate the failure mode rather than to carry evidential weight; representative cell-level numbers are in Table 2, Appendix D.

B Calibration beyond ECE, and cell-level multiplicity

ECE depends on binning and on sample composition, so we report two binning-free proper scoring rules alongside it. On Mistral × MATH (750 records, accuracy 0.648), the model's retrospective stated confidence gives Brier 0.262 and NLL 1.470 against a 10-bin ECE of 0.271. P(True) on the same records gives Brier 0.386: better ordered than stated confidence, worse calibrated in absolute terms. The steering results in Table 1 are reported as ECE because the steered runs stored per-gain aggregates rather than per-record confidences; recomputing Brier and NLL at each gain requires re-running the intervention and is the first thing we would add.

The cell-level bootstrap counts in Table 2 are uncorrected. With 16 cells, some individually significant results are expected under the null, so the cell counts should be read as descriptive. The aggregate pattern is the claim that carries weight: the effect is positive in 12 of 16 cells with one tie, significantly positive in 10, and never significantly negative. A hierarchical model pooling across cells with model and domain as random effects is the right test and is not yet run.

C Alternative explanations for the probe signal

A probe could key on properties of failed derivations rather than on a dedicated metacognitive variable: low token probability, hedging language, or, on MATH, an early-layer channel next to the answer text. Our surface-features baseline (answer length, digit count, line count, word count, brace count) bounds this without closing it, and the bound is domain-dependent: the probe adds +0.21 over surface on MMLU-Pro but only +0.07 on MATH and −0.11 on geometry, where a construction that fails to compile is visibly malformed. Two results argue against a pure surface account on the domains where the signal is strong. A direction fitted on MMLU-Pro, where surface reaches only 0.54 against a probe of 0.75, still governs MATH's self-report under ablation ($0.880 \to 0.709$). And the pre-attempt read, taken where the activation is byte-identical across a question's attempts, cannot recover the post-attempt direction on MMLU-Pro (0.490, chance), which a surface-of-the-question account would predict it could. A linear probe at one token also undercounts nonlinear or multi-token signal, so the reported numbers are a floor.

D Per-cell numbers and specificity controls

E Checkpoints

All models are public Hugging Face releases, run in bfloat16 without quantization, greedy decoding: `mistralai/Mistral-Small-24B-Instruct-2501`, `Qwen/Qwen3.6-27B`, `zai-org/GLM-4.7-Flash`, `google/gemma-4-26B-A4B-it`; the dense within-family control for the lens is `Qwen/Qwen2.5-14B-Instruct`.

Table 2: Post-attempt correctness discrimination (AUROC). Left: stated confidence vs. the probe in representative cells; overall the probe wins 12 of 16 with one tie, mean +0.09, with a paired bootstrap significantly positive in 10 of 16 and never significantly negative, uncorrected at the cell level (Appendix B). Right: Mistral specificity controls. "pre" is the last prompt token before the attempt; "post" is after the attempt; "resid." removes the pre score; "post→pre" applies that fitted post readout at the pre-attempt site, where only difficulty is visible.

model × domain	**stated**	**probe**
Gemma-4 × MATH	0.57	**0.78**
Qwen3.6 × MMLU-Pro	0.67	**0.84**
GLM × MMLU-Pro	0.67	**0.85**
Mistral × MATH	0.79	**0.83**

Mistral	**pre**	**post**	**resid.**	**post→pre**
MMLU-Pro	0.568	0.754	0.751	0.490
MATH	0.792	0.834	0.701	0.624
GPQA	0.593	0.572	0.544	0.479
GeoGen	0.770	0.676	0.505	0.648